

# An inter-city optical clock network for precision metrology and relativistic geodesy

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## Abstract

A redefinition of the SI second relies on optical frequency ratios that different laboratories measure independently and that agree to within  $5 \times 10^{-18}$ . Many optical clocks report self-evaluated uncertainties meeting this level, but no comparison between clocks in different cities has yet reached it. Here we close this gap with an inter-city, cross-species clock network: an IAPMST ytterbium clock in Wuhan and a USTC strontium clock in Shanghai, about 700 km apart, are linked through a 1350 km fibre link and compared coherently for 58 days (280 h). Treating the tidal gravitational-redshift term as an uncertainty item, we obtain the Yb/Sr frequency ratio ~~1.207 507 039 343 337 721 3(24)~~1.207 507 039 343 337 721 3(23), with a statistical uncertainty of  $0.75 \times 10^{-18}$  and a total uncertainty of  $2 \times 10^{-18}$ , whose central value differs from the latest NIST–JILA result by less than  $5 \times 10^{-18}$ . On the same data the tidal redshift modulation is resolved at  $6.4\sigma$ ; removing it lowers the reduced chi-square from 5.42 to 3.70 (32%). The two clocks also read the time-varying gravitational potential over the real 700 km baseline, resolving a tidal geopotential difference at centimetre-level equivalent height. Such a network thus both supports the redefinition of the SI second and senses the time-varying gravitational potential.

## 1 Introduction

Optical clocks now measure frequency well enough to sense the gravitational redshift over a height difference of a centimetre<sup>1</sup>, with self-evaluated fractional uncertainties below  $1 \times 10^{-18}$ <sup>2,3</sup> and two-clock instabilities approaching  $10^{-18}$  within an hour<sup>4</sup>, and their performance is driving two goals at once: redefining the SI second on an optical transition and reading the Earth’s gravitational potential from clock comparisons<sup>5,6</sup>. Both goals, however, depend on a capability that a single location cannot provide. The best clocks are not transportable, so their claimed accuracy can only be confirmed by comparing them with one another over long distances. This has motivated the concept of a clock *network*, in which separate clocks are linked into a single cooperative instrument<sup>7,8</sup>.

Fibre links have made such comparisons possible. Ultrastable light can now be transferred over hundreds to thousands of kilometres with added noise below the level of the clocks themselves<sup>9,10</sup>, enabling inter-city links<sup>11</sup>, multi-species ratio measurements<sup>12</sup>, and, most recently, a fibre network spanning several countries<sup>13</sup>; transportable clocks, satellite links and free-space dissemination have extended comparisons further<sup>14–16</sup>. Beyond metrology, such networks also probe fundamental physics, from gravitational waves<sup>17</sup> to dark matter and topological defects<sup>18</sup>. Yet the comparisons reported so far fall short of what a redefined second requires. The CCTF roadmap calls for optical frequency ratios measured independently in different institutes and agreeing at the few-parts-in- $10^{18}$  level, whose closure also validates the estimated uncertainties<sup>5,19</sup>: the only cross-species ratio reported at this level was measured over a short baseline between two clock laboratories in the same city<sup>20</sup>, and the widest networks have not yet reached  $5 \times 10^{-18}$ <sup>15</sup>.

It therefore remains unclear whether two clocks of different species, built at different institutions and separated by hundreds of kilometres of fibre, can agree at the level a redefined second demands. Here we show that they can. We link an ytterbium lattice clock at IAPMST in Wuhan



Figure 1: Experimental setup. The IAPMST Yb lattice clock in Wuhan and the USTC Sr lattice clock in Shanghai are linked by a phase-stabilised 1550 nm fibre of single-pass optical length  $\approx 1350$  km, with loop-back purification at each repeater; the out-of-loop beat note between the received carrier and the local comb is the raw observable.

to a strontium lattice clock at USTC in Shanghai, about 700 km apart, through a 1350 km phase-stabilised fibre, and operate them coherently for 58 days. Treating the tidal gravitational-redshift term as an uncertainty item, we obtain a Yb/Sr ratio consistent with the existing determinations and detect the tidal modulation of the ratio at  $6.4\sigma$ .

## 2 Results

### 2.1 Network and observation campaign

The experiment compares a Yb lattice clock at IAPMST in Wuhan<sup>21</sup> with a Sr lattice clock at USTC in Shanghai<sup>22</sup>, two institutions and two clock species about 700 km apart. They are joined by a phase-stabilised 1550 nm fibre link of single-pass optical length  $\approx 1350$  km, deployed on paired strands within a single cable duct of a telecommunications operator and organised as a cascade of relay stations, each carrying a bidirectional erbium-doped fibre amplifier and a fibre-noise-cancellation loop. Loop-back purification keeps the link instability below the clock noise beyond  $\sim 100$  s (Fig. 1; Methods). The transfer architecture follows the cascaded, bias-corrected optical-frequency-dissemination design demonstrated on a 2067 km field link<sup>10</sup>; the present link is shorter, which relaxes the accumulated noise. The raw observable is the out-of-loop beat note between the received carrier and the local comb, sampled at 1 s.

Over a 58-day campaign (2026-06-29 to 2026-08-26), 17 uninterrupted segments yield 1 008 912 valid one-second samples (280 h, a duty cycle of about 20%), gated by the simultaneous lock of both clocks and the comb chain rather than by the link. The two clocks were restarted independently between segments in the first campaign, so the ratio centre provides a direct reproducibility check and shows no measurable bias across restarts. The static geopotential difference,  $\Delta W = 280.042 \pm 0.085 \text{ m}^2/\text{s}^2$  (Wuhan higher), is fixed by first-order spirit levelling and applied as a constant correction (Methods); its uncertainty enters the ratio budget as the  $1 \times 10^{-18}$  static-redshift term (Table 1). For the metrological result we treat the time-varying tidal geopotential difference,  $\Delta W(t) = W_{\text{WUHN}}(t) - W_{\text{SHAN}}(t)$ , as an uncertainty item (Sec. 2.4), taking its waveform from a professional 30 s series that combines solid-Earth tide and ocean loading.

Table 1: Corrections and uncertainty terms for the Yb/Sr ratio (fractional frequency units). “Correction” is the applied shift to the central value; “Uncertainty” is the term’s contribution to the budget, combined in quadrature. The tidal term is carried as an uncertainty item (no correction applied; Sec. 2.4). The total is the quadrature sum of the listed terms.

| Source                        | Correction             | Uncertainty            |
|-------------------------------|------------------------|------------------------|
| Sr-clock systematic           | $9.2 \times 10^{-19}$  | $9.2 \times 10^{-19}$  |
| Yb-clock systematic           | $1.1 \times 10^{-18}$  | $1.1 \times 10^{-18}$  |
| Static gravitational redshift | $-3.1 \times 10^{-15}$ | $1 \times 10^{-18}$    |
| Fibre transfer                | —                      | $< 1 \times 10^{-19}$  |
| Comb transfer                 | —                      | $< 1 \times 10^{-19}$  |
| Statistical (Bayesian)        | —                      | $0.75 \times 10^{-18}$ |
| Tidal term (item)             | —                      | $< 5 \times 10^{-19}$  |
| Total                         | —                      | $2 \times 10^{-18}$    |

## 2.2 Clock-ratio determination

Combining the per-segment ratios with a random-effect model gives the uncorrected Yb/Sr frequency ratio

$$\text{Yb/Sr} = 1.207\,507\,039\,343\,337\,721\,3(23), \quad (1)$$

with a Bayesian statistical uncertainty of  $0.75 \times 10^{-18}$  from the clock statistical noise alone. The parenthetical (23) is the display precision of the value string, not the statistical term. Removing the static geopotential shifts the centre by only  $0.2 \times 10^{-18}$  relative to the experiment-side tidal-corrected value, and an independent re-derivation from the raw beat files reproduces the per-segment ratios exactly (Methods).

Table 1 lists the uncertainty terms. The Yb-clock evaluation<sup>21</sup> ( $1.1 \times 10^{-18}$ ) and the static gravitational redshift ( $1 \times 10^{-18}$ ) dominate; the Sr-clock term<sup>22</sup> is  $9.2 \times 10^{-19}$ , the fibre and comb terms are each below  $1 \times 10^{-19}$ , and the tidal term is carried as an uncertainty item of  $< 5 \times 10^{-19}$ , the size of the theoretical compensation. Combined in quadrature these terms give a total fractional uncertainty of  $2 \times 10^{-18}$ . The clock-systematic terms are the published evaluations of the two clocks<sup>21;22</sup>, and the tidal-model contribution is a theoretical bound; a full point-by-point systematics evaluation remains in progress.

Figure 2 shows the clock-ratio data behind Eq. (1): the 17 per-segment ratios  $R_i$  and the segment-to-segment stability. In terms of the deviations  $y_i = R_i/R_{\text{seg1}} - 1$ , the duration-weighted mean is  $-0.482 \times 10^{-18}$  (arithmetic mean  $+0.102 \times 10^{-18}$ ) and the segment-to-segment scatter is  $\sigma(y_i) = 2.98 \times 10^{-18}$ .

## 2.3 Tidal gravitational-redshift detection

The tide modulates the comparison frequency through the gravitational redshift,  $\Delta f/f/f = \Delta W(t)/c^2$ , with a segment-averaged rms amplitude of  $3.84 \times 10^{-18}$  (median; range  $0.86\text{--}6.56 \times 10^{-18}$  across the 17 segments).

After both the beat and a tidal template are passed through a 1200 s triangular window and demeaned, each segment is fitted with  $A = \sum(\text{tide} \cdot \text{beat}) / \sum \text{tide}^2$ . The template is normalised to the 1550 nm transfer frequency  $F_{1550} = 193.40$  THz, not to the ratio-conversion coefficient  $1/\text{COEF} \approx 233.53$  THz; the two differ by the Yb/Sr factor 1.2075, so the choice of reference sets the amplitude scale and must be accounted for before comparing  $A$  with other tidal analyses (Methods). A single segment reaches only  $\text{SNR} \approx 0.28$ , so no segment is individually significant (Fig. 3); across the 17 segments, however, the detection is significant by three independent combinations: 14/17 segments share the same sign ( $p = 0.013$ ), the Stouffer sign-agnostic  $|z| =$

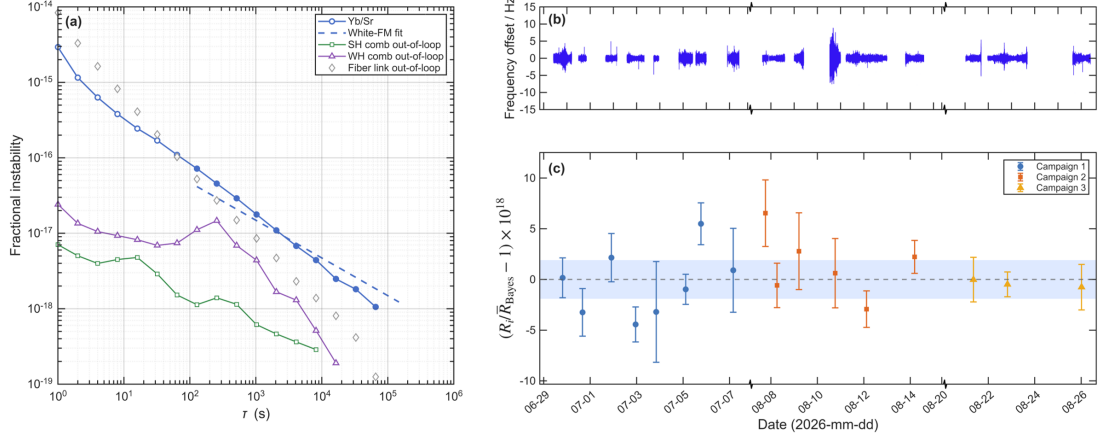


Figure 2: Clock-ratio data. The segment stability and the 17 per-segment Yb/Sr ratios  $R_i$  give the deviation  $y_i = R_i/R_{\text{seg1}} - 1$  whose duration-weighted mean is  $-0.482 \times 10^{-18}$  and whose segment-to-segment scatter is  $\sigma(y_i) = 2.98 \times 10^{-18}$ .

5.87 ( $p = 4.3 \times 10^{-9}$ ), and Fisher  $p = 2.2 \times 10^{-6}$ . The precision-weighted amplitude,

$$A = -0.5397 \pm 0.0843 \quad (\approx 6.4\sigma \text{ from zero}), \quad (2)$$

is detected with the correct sign at roughly half the theoretical magnitude. At the segment-mean level, the ratio deviation  $y_i$  is positively correlated with the tidal shift  $\Delta f/f$  (Pearson  $r = +0.518$ ,  $p = 0.033$ ).

## 2.4 Tidal effect and its compensation

At the full theoretical response ( $A = -1$ , complete trust in the tidal data and in general relativity) the tide would shift the ratio by  $0.44 \times 10^{-18}$ : in fractional-frequency terms the duration-weighted centre would move from 1.207 507 039 343 337 720 369 6 (raw) to 1.207 507 039 343 337 720 810 8. That is small against the  $0.75 \times 10^{-18}$  statistical term but comparable to the leading systematics, so we do not correct the central value; the tidal term enters the budget as an uncertainty item of  $< 5 \times 10^{-19}$ , its size set by that full-response shift (Table 1).

We asked whether an empirical response below the theoretical  $A = -1$  is justified. The measured amplitude is  $A = -0.54 \pm 0.08$  (Sec. 2.3), roughly half the theoretical magnitude, but no independent check establishes the fitted value as the correct response coefficient: the fit reproduces the waveform, not the hardware phase that fixes the sign convention. We therefore do not adopt it as a compensation.

A half-strength response admits several physical readings, none of which the present data can single out. The professional tidal series combines solid-Earth tide with ocean loading, yet the loading term over a 700 km inter-city baseline depends on the local coastline and sedimentary structure and on groundwater and hydrological loading that a global model resolves only coarsely; a phase- or amplitude-mismatched loading component would reduce the coherent response without removing it. Alternatively, the two clocks sense slightly different local geopotentials, and any common-mode environmental coupling along the fibre path that partially tracks the tide would also dilute the fitted amplitude. Because these mechanisms are degenerate at the present sensitivity, we keep the uncorrected centre as the headline ratio and report the amplitude as an effect estimate rather than a calibration, leaving a per-component tidal model as the natural route to a full-strength response.

Compensating at the fitted  $A = -0.54$  would shift the duration-weighted centre by  $+0.24 \times 10^{-18}$ , half the  $+0.44 \times 10^{-18}$  of the full theoretical response, both below the present statistical

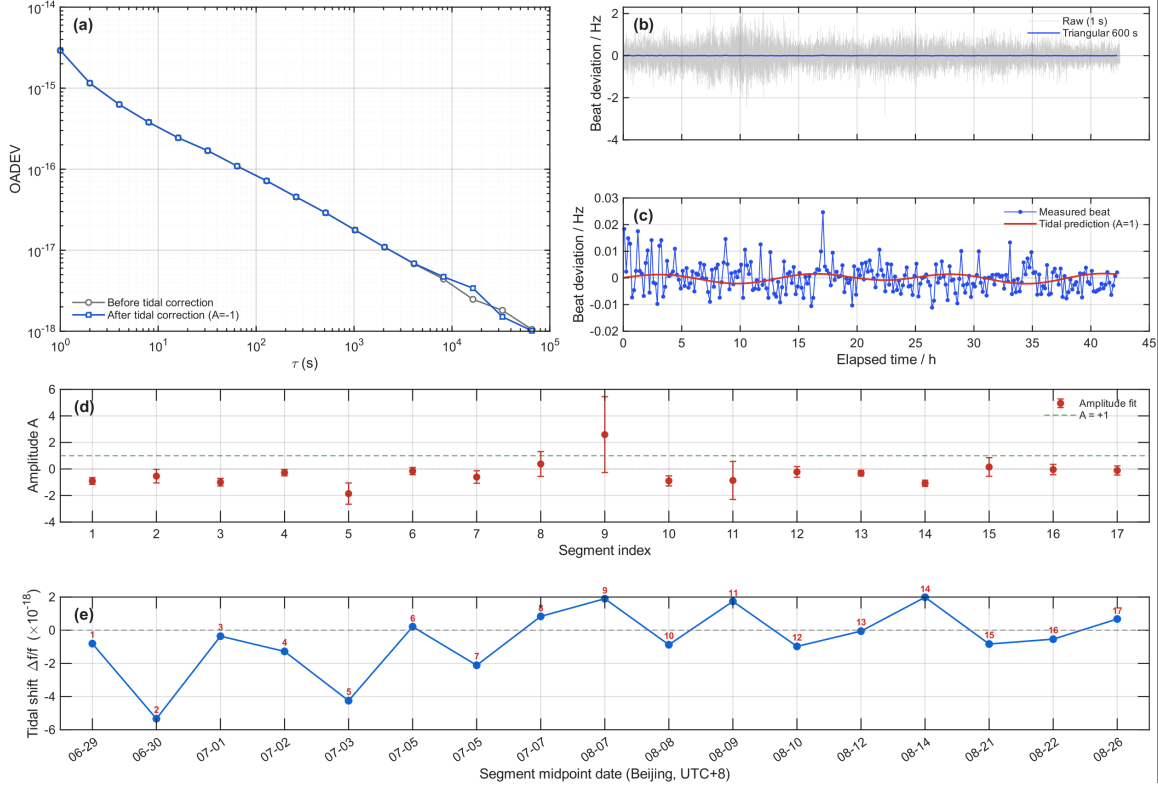


Figure 3: Tidal influence and its correlation. The tidal modulation of the comparison frequency is resolved at  $6.4\sigma$  across the 17 segments (combined amplitude  $A = -0.54 \pm 0.08$ , at about half the theoretical magnitude), and at the segment-mean level the ratio deviation  $y_i$  is positively correlated with the tidal shift  $\Delta f/f$  (Pearson  $r = +0.518$ ).

uncertainty. The empirical compensation would lower the reduced chi-square from 5.42 (raw) to 4.56 and the segment-to-segment dispersion  $\sigma(y_i)$  from  $2.980 \times 10^{-18}$  to  $2.596 \times 10^{-18}$  (13%;  $12.9\% \pm 11.0\%$  by jackknife), while the full theoretical response gives a larger  $\chi^2_{\text{red}}$  reduction ( $5.42 \rightarrow 3.70$ ) but a slightly *higher* long-term instability; neither removes the residual at long averaging times. Because neither the statistics nor the sign convention validates a correction, the amplitude is reported as an effect estimate rather than adopted (Supplementary Information).

### 3 Discussion

Two features set this network apart. It is inter-city and cross-species: two clock species<sup>21;22</sup>, at two institutions and two cities, joined only by the fibre link and the combs, yet yielding a ratio that agrees with the leading independent determinations at the  $5 \times 10^{-18}$  level without any transport of the clocks. And the same data set senses the time-varying gravitational potential, the  $0.75 \times 10^{-18}$  statistical floor corresponding to an equivalent height of about 0.7 cm and the tidal modulation being resolved at  $6.4\sigma$  across the baseline.

The comparison places the Yb/Sr ratio on the map of independent determinations (Table 2, Fig. 4). Our uncorrected centre lies  $-1.7 \times 10^{-18}$  below the NIST–JILA value<sup>20</sup> and  $+3.3 \times 10^{-18}$  above the European-network value<sup>13</sup>. The NIST–JILA determination carries a total uncertainty at the few-parts-in- $10^{18}$  level, so our difference from it is below the  $5 \times 10^{-18}$  target; the European-network central value is quoted for completeness rather than as an independent agreement test, since its statistical uncertainty is substantially larger, and the near-coincidence of the three values should not be read as a three-way agreement at  $5 \times 10^{-18}$ . The choice of uncorrected or tidal-corrected centre and of Bayesian or duration-weighted averaging moves any of these deviations

Table 2: Yb/Sr frequency-ratio determinations. Deviations are given relative to the present work (uncorrected centre, Sec. 2.2), in units of  $10^{-18}$ .

| Determination    | Yb/Sr                           | Deviation | Ref.          |
|------------------|---------------------------------|-----------|---------------|
| NIST-JILA        | 1.207 507 039 343 337 7230(37)  | -1.7      | <sup>20</sup> |
| This work        | 1.207 507 039 343 337 721 3(23) | 0         | —             |
| European network | 1.207 507 039 343 337 718(32)   | +3.3      | <sup>13</sup> |

by at most  $\sim 0.7 \times 10^{-18}$ , below the scale of interest.

The immediate metrological value of this work is as an independent check on the Yb/Sr ratio that the redefinition of the SI second will rest on. That redefinition calls for a matrix of ratios measured by different laboratories and mutually agreeing at the few-parts-in- $10^{18}$  level; for Yb/Sr only a single determination to date meets the  $5 \times 10^{-18}$  target<sup>20</sup>, and it differs markedly from earlier reports. The present result adds a second, independently obtained entry to that matrix, from a third institution with a different pair of clocks and separate hardware and analysis, and so tests the redefinition on evidence the single-campus determination cannot provide. We stress that this is a conservative claim: the  $0.75 \times 10^{-18}$  figure is a statistical floor, and we report a ratio consistent with the existing determinations rather than a new most-precise value.

The same data set also bounds what the comparison can support. The bottleneck is no longer the link or the comb, whose contributions are each below  $1 \times 10^{-19}$  (Table 1), but the segment-to-segment scatter; closing the gap between the  $0.75 \times 10^{-18}$  statistical floor and the  $\sim 5 \times 10^{-18}$  spread between independent determinations therefore requires progress on the systematic side, in the clock evaluations<sup>21;22</sup> and in joint geodetic levelling, rather than longer integration alone. Two further limits should be read with the result: the per-segment uncertainties  $u_i$  are unreliable (an OADEV extrapolation that underestimates by roughly 30–40%, with anomalous segments 5, 6 and 16), so the combined values are for method comparison only; and the negative sign of the fitted amplitude is a specified response convention rather than a resolution of the hardware-polarity question.

## 4 Outlook

Beyond metrology, the same comparison is a candidate sensor for chronometric geodesy: the ratio and the geopotential difference are read from one measurement, and the tidal analysis here demonstrates the required sensitivity on a single baseline. A genuine relativistic coordinate reference frame, however, would require several nodes with closing loops, geodetic ties, a frame convention and continuous operation; a two-node, single-baseline,  $\sim 20\%$  duty-cycle experiment is the elementary link of such a network, not the network itself. Extending this prototype node pair to a multi-node network, with improved link and comb stability and dedicated joint levelling, is the natural next step.

Beyond geodesy, the same network reaches into fundamental physics, where clock comparisons constrain dark-matter couplings<sup>23;24</sup> and test Lorentz symmetry<sup>25</sup>; realizing that reach on an inter-city baseline is an open question motivated by the present sensitivity.

## 5 Conclusion

We have demonstrated an inter-city, cross-species optical-clock network: an IAPMST Yb clock in Wuhan<sup>21</sup> and a USTC Sr clock in Shanghai<sup>22</sup>, about 700 km apart and linked by a phase-stabilised fibre, operated coherently for 58 days. Combining 17 segments with a Bayesian random-effect model gives an uncorrected Yb/Sr ratio of 1.207 507 039 343 337 721 3(23) with a total uncertainty of  $2 \times 10^{-18}$ , consistent with the existing determinations at the  $5 \times 10^{-18}$  level; the

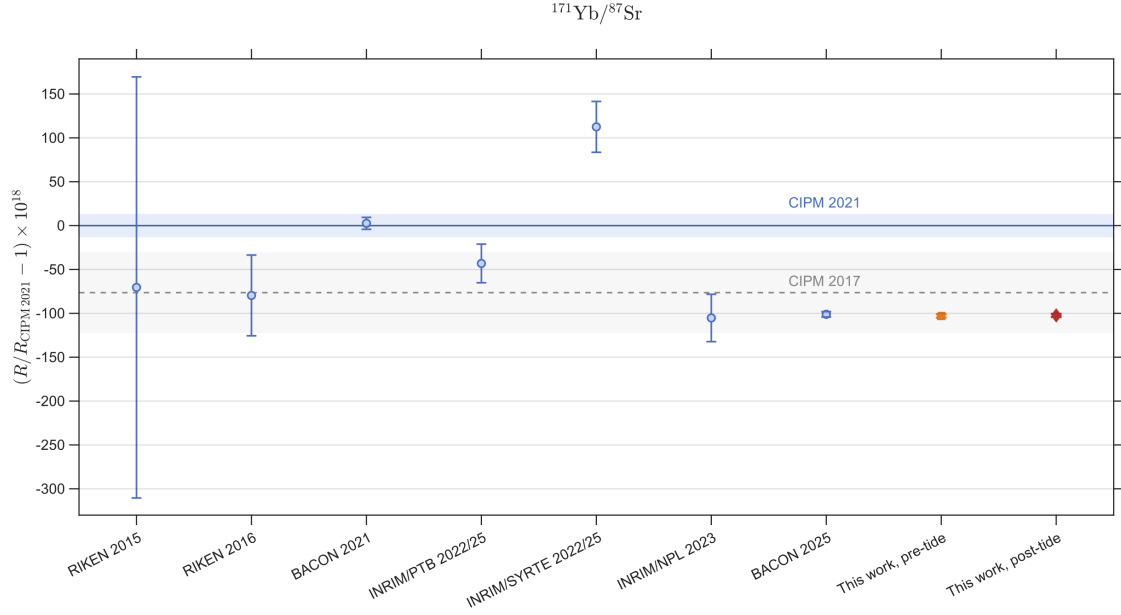


Figure 4: Comparison with previous determinations of the Yb/Sr ratio. The combined centres of this work, with the experiment-side WLS and NIST reference values, place the present result alongside the existing independent determinations.

tidal gravitational-redshift modulation of the same data is resolved at  $6.4\sigma$  and carried as an uncertainty item. Two capabilities follow. The comparison supplies an independently obtained entry for the Yb/Sr ratio that a redefined SI second will rest on. And, for the first time over a real 700 km baseline rather than a single campus, the two independent clocks resolve the time-varying gravitational potential at a centimetre-level equivalent height, casting the inter-city clock network as a prototype relativistic-geodesy sensor as well as a metrological one.

## 6 Methods

We describe the measurement and its analysis in three groups: the apparatus and the link (§“Experimental setup and frequency transfer”), the data set (§“Observation campaign and per-segment data”), and the analysis chain that turns the raw beat into the reported ratio—the ratio inversion, the systematic and geopotential corrections, the tidal treatment, and the statistical combination of the per-segment results.

### 6.1 Experimental setup and frequency transfer

The main text keeps only the conceptual frequency chain; here we record the symbol conventions that make the inversion reproducible. The observable is the beat between the received 1550 nm transfer carrier and the local Yb-referenced comb; every frequency in the chain is derived from the comb repetition rate  $f_{\text{rep}} = f_{\text{ref}} \times \text{DIV20} = 200 \text{ MHz}$ .

The fibre link itself is built on the cascaded, bias-corrected optical-frequency-dissemination architecture of Ref.<sup>10</sup>: the route is divided into segments, each terminated by a relay station that carries a bidirectional erbium-doped fibre amplifier and a fibre-noise-cancellation loop, with the round-trip phase noise measured in the radio-frequency domain. Deployed on paired strands within a single telecommunications cable duct, the present Shanghai–Wuhan route is shorter than the 2067 km field link of Ref.<sup>10</sup>, so the accumulated link noise is correspondingly lower; the relay stations used here operate without the dedicated sub-hertz noise-purification stage that the

Table 3: AOM frequency offsets and their roles in the Shanghai–Wuhan link and comb chains (from the experiment-side supplementary material). FNC denotes a fibre-noise-cancellation loop; BBR a blackbody-radiation correction loop.

| Index | Offset and role                                          |
|-------|----------------------------------------------------------|
| 1     | +80 MHz, fixed                                           |
| 2     | +80.274 MHz, FNC + closed-loop feedback                  |
| 3     | +80 MHz, FNC + BBR feedback                              |
| 4     | −110 MHz, FNC feedback                                   |
| 5     | +50 MHz, FNC feedback                                    |
| 6     | +236 MHz, clock-laser–cavity offset; cavity feed-forward |
| 7     | −80 MHz, FNC feedback                                    |
| 8     | +155 MHz, frequency compensation and correction          |
| 9     | −80 MHz, FNC feedback                                    |
| 10    | +105 MHz, frequency compensation and correction          |
| 11    | +80 MHz, FNC feedback                                    |
| 12    | −140 MHz, clock interrogation control                    |
| a     | +50 MHz, feedback, fibre transfer FNC                    |
| b     | −40 MHz, fixed                                           |

longer link requires.

At the Shanghai end the Sr clock at 698 nm is transferred to a 1397 nm carrier; at the Wuhan end a second comb links the received 1550 nm carrier to the Yb clock at 578 nm through a 1156 nm fundamental beam. The clock-laser frequency relations are

$$f_{\text{comb}}^{1397} = \frac{1}{2} (1 + a) f_{\text{Sr}}, \quad a = a_{\text{dens}} + a_{\text{lat}} + a_{Z^2} + a_{\text{BGC}}, \quad (3)$$

$$f_{\text{comb}}^{1156} = \frac{1}{2} (1 + b) f_{\text{Yb}}, \quad (4)$$

where  $a$  collects the Sr-side systematic corrections<sup>22</sup> and  $b$  the Yb-side correction<sup>21</sup> ( $b_{\text{Yb}} = -1.7 \times 10^{-18}$ ; the internal components are listed below and in the repository). Table 3 gives the AOM frequency offsets and their roles in the link and comb chains.

The static geopotential difference between the two clocks,  $\Delta W = 280.04164 \text{ m}^2/\text{s}^2$ , is applied at the ratio level as a constant correction  $\delta_g = \Delta W/c^2 = -3.1159 \times 10^{-15}$  (fixed by the levelling described below):  $(\text{Yb}/\text{Sr})_{\text{corr}} = (\text{Yb}/\text{Sr})_{\text{raw}} \times (1 + \delta_g)$ .

## 6.2 Observation campaign and per-segment data

The comparison spans the 58-day campaign described in the main text. Table 4 lists the 17 uninterrupted segments, their retained sample counts  $n_i$  (after endpoint trimming) and measurement campaign; the retained total is  $\sum_i n_i = 1\,008\,912$  one-second samples. During campaign 1 the Yb and Sr clocks were restarted independently between some segments (Yb after segments 4–6; Sr after segments 7–8), which lets the reproducibility of the ratio be checked directly.

## 6.3 Beat-to-ratio inversion

We take  $b_i(t)$  as the longest valid beat sequence in segment  $i$  and  $m$  as the global median of the plausible beat values, so the segment beat deviation is

$$\overline{\delta b_i} = \overline{b_i} - m. \quad (5)$$

The Sr-side systematic shift  $a_i$  and the Yb-side shift  $b_{\text{Yb}}$  combine into the Dr coefficient, and the segment Yb/Sr ratio follows from the full inversion

$$R_i = [\text{ratio\_base}_i + \text{Dr}_i]^{-1} \times (1 + \delta_g), \quad \delta_g = -3.1159 \times 10^{-15}, \quad (6)$$

Table 4: The 17 uninterrupted segments: index, retained samples  $n_i$  (endpoint-trimmed), and Beijing-time start.

| Segment | $n_i$ (s) | Start (Beijing)     |
|---------|-----------|---------------------|
| 1       | 68 009    | 2026-06-29 10:06:28 |
| 2       | 29 481    | 2026-06-30 12:00:01 |
| 3       | 42 891    | 2026-07-01 15:58:41 |
| 4       | 64 280    | 2026-07-02 14:00:01 |
| 5       | 19 532    | 2026-07-03 17:34:22 |
| 6       | 52 140    | 2026-07-04 19:30:46 |
| 7       | 39 526    | 2026-07-05 13:00:01 |
| 8       | 43 565    | 2026-07-06 21:45:16 |
| 9       | 22 496    | 2026-08-07 15:15:00 |
| 10      | 57 938    | 2026-08-07 22:15:00 |
| 11      | 35 824    | 2026-08-09 00:00:00 |
| 12      | 40 346    | 2026-08-10 12:47:06 |
| 13      | 152 994   | 2026-08-11 05:30:00 |
| 14      | 68 618    | 2026-08-13 18:56:58 |
| 15      | 57 649    | 2026-08-21 00:40:03 |
| 16      | 147 640   | 2026-08-21 23:20:01 |
| 17      | 65 983    | 2026-08-25 15:09:41 |
| Total   | 1 008 912 |                     |

where  $\delta_g$  is the static gravitational correction. All ratio arithmetic is carried out at 80-decimal precision (`Decimal80`, mirroring MATLAB `vpa(...,80)`); a float64 absolute precision of about  $2.6 \times 10^{-16}$  would destroy differences at the  $10^{-18}$  level. The duration-weighted centre of the whole experiment is

$$R_{\text{duration}} = \frac{\sum_i n_i R_i}{\sum_i n_i}, \quad \sum_i n_i = 1\,008\,912. \quad (7)$$

#### 6.4 Systematic frequency shifts

The ratio inversion removes the known systematic shifts of both clocks through the coefficients  $a_i$  (Sr side) and  $b$  (Yb side). The Yb-side correction is a fixed value,  $b_{\text{Yb}} = (5.3 - 7.0) \times 10^{-18} = -1.7 \times 10^{-18}$ , entering as  $\text{coef}_{1156} = (1 + b_{\text{Yb}})/2$ . The Sr-side correction is segment-dependent and is the sum of five terms,

$$a_i = a_{\text{rou},i} + a_{\text{AC},i} + a_{\text{SM},i} + a_{\text{air},i} + a_{\text{BBR},i}, \quad \text{coef}_{1397,i} = \frac{1}{2}(1 + a_i), \quad (8)$$

where  $a_{\text{rou}}$  is the lattice light-shift,  $a_{\text{AC}}$  the ac-Stark shift ( $+7.444 \times 10^{-18}$  in the first campaign,  $+8.929 \times 10^{-18}$  in the second),  $a_{\text{SM}}$  the second-order Zeeman shift ( $\approx -1.78 \times 10^{-16}$  in the first campaign;  $-8.925 \times 10^{-17}$  in segment 9 and zero in segments 10–14),  $a_{\text{air}}$  the air-refractivity correction ( $-6.7 \times 10^{-19}$ ) and  $a_{\text{BBR}}$  the blackbody-radiation shift (zero). Segments 15–17 reuse the constant  $a_i$  of segments 12–14. The per-segment values are stored in the repository (`clock/params.json`); the anomalous `a_SM` of segment 9 is retained as delivered, without adjustment. The component evaluations follow the published assessments of the two clocks<sup>21;22</sup>.

#### 6.5 Levelling determination of the geopotential difference

The static gravitational-redshift correction  $\delta_g$  entering the ratio inversion is fixed by the geopotential difference between the two clocks' atomic transition positions, determined by first-order

Table 5: Partial geopotential differences between the Yb (Wuhan) and Sr (Shanghai) clocks, in  $\text{m}^2/\text{s}^2$ .

| Segment                                           | $\Delta W$ ( $\text{m}^2/\text{s}^2$ ) |
|---------------------------------------------------|----------------------------------------|
| Yb atom $\rightarrow$ Yb optical platform         | +2.65399                               |
| Yb platform $\rightarrow$ benchmark WS-03         | −14.51715                              |
| Benchmark WS-03 $\rightarrow$ benchmark WS-01     | +296.33022                             |
| Benchmark WS-01 $\rightarrow$ Sr optical platform | −1.47246                               |
| Sr platform $\rightarrow$ Sr atom                 | −2.95296                               |
| Yb atom $\rightarrow$ Sr atom (total)             | +280.04164                             |

spirit levelling together with simultaneous gravimetry. A levelling line of about 1113 km was run between the Wuhan benchmark WS-03 (at IAPMST) and the Shanghai benchmark WS-01 (at USTC), comprising six first-order routes through Ezhou, Huangshi, Huanggang, Lu’an, Hefei, Wuhu, Guangde and Suzhou. Levelling was performed with digital levels (Sokkia SDL1X, Trimble DiNi03) and bar-coded invar staffs, with sight lengths below 30 m and a forward-back sight-length difference below 1.0 m; gravity was observed at every levelling point with two relative gravimeters. The levelled normal heights were converted to orthometric heights for the geopotential integration, and the geopotential difference between the two benchmarks follows from  $\Delta W_{AB} = -\sum_i g_i \Delta H_i$ , where  $g_i$  is the mean gravity and  $\Delta H_i$  the orthometric height difference of the  $i$ -th segment.

Two further segments connect each benchmark to the corresponding atomic transition position: the benchmark-to-optical-platform height difference was obtained by trigonometrical heighting (total station), and the platform-to-atom segment from the clock design parameters. The five partial geopotential differences are listed in Table 5 and sum to  $\Delta W = 280.04164 \text{ m}^2/\text{s}^2$  between the two atomic transition positions. The levelling precision, expressed as the accidental mean error per kilometre, is 0.255 mm, better than the 0.45 mm limit for first-order levelling. Propagating the levelling, trigonometrical-heighting and platform-to-atom terms yields a total height precision of  $\pm 0.0086 \text{ m}$ , and hence a geopotential-difference uncertainty of  $\pm 0.085 \text{ m}^2/\text{s}^2$ .

## 6.6 Tidal analysis

The tidal beat template is formed from the professional 30 s geopotential-difference series, linearly interpolated to the retained beat timestamps after shifting the Beijing-time beat windows to UTC. It is normalised to the 1550 nm transfer frequency,  $h(t) = F_{1550} \Delta W(t)/c^2$  with  $F_{1550} = N_{1550}^{\text{WH}} f_{\text{rep}} = 193.3992 \text{ THz}$ , *not* to  $1/\text{COEF}$ , which would be larger by the Yb/Sr factor  $\approx 1.2075$  and would bias  $A$  low by that factor. Within each segment the beat and the template are passed through a 1200 s triangular window (600 s step) and, after demeaning, fitted with  $\text{beat} = A \text{tide} + \text{noise}$ ,  $A = \sum(\text{tide} \cdot \text{beat}) / \sum \text{tide}^2$ . The resulting segment-mean correlation between  $y_i$  and  $\Delta f/f$  is shown in Fig. 5. We combine the per-segment amplitudes by precision weighting, together with the sign-consistency (binomial), Stouffer (sign-agnostic  $|z|$ ) and Fisher combinations.

For the fixed-coefficient scenarios we apply the correction to the beat before re-inverting the ratio,  $b_{\text{corr}}(t) = b_{\text{raw}}(t) - A h(t)$  with  $A = -1$  (theory) or  $A = -0.54$  (empirical); the ratio chain is then re-inverted and the four combination methods re-applied. The direction self-check and the dispersion reduction are reported in the Results. Because the two amplitudes are fixed rather than fitted, and the sign follows a specified response convention (see Discussion), we report these scenarios as effect estimates and keep the uncorrected centre as the headline ratio.

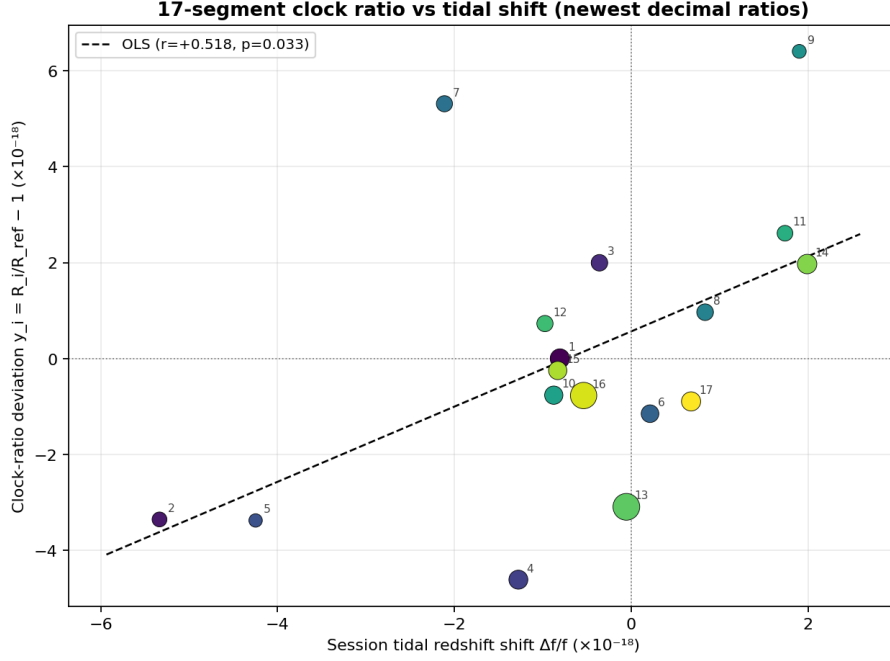


Figure 5: Segment-mean correlation. Clock-ratio deviation  $y_i = R_i/R_{\text{seg1}} - 1$  against the session tidal shift  $\Delta f/f$  for the 17 segments; the ordinary least-squares fit gives the positive correlation reported in the Results.

## 6.7 Per-segment uncertainties

Each segment is assigned a statistical uncertainty from the overlapping Allan deviation (OADEV) of its fractional frequency series  $\delta f/F_{1550}$ . Over the white-frequency-noise range  $128\text{ s} \leq \tau \leq T/4$  the OADEV follows  $\sigma_y(\tau) \propto \tau^{-1/2}$ ; a linear fit of  $\log \sigma_y$  against  $\log \tau$  is extrapolated to the segment length  $T$  to give  $\sigma_y(T)$ , and

$$u_i = \sigma_y(T) R_i \quad (9)$$

is the absolute segment uncertainty on  $R_i$ . A flicker floor beyond  $\tau \sim 4096\text{ s}$  makes this extrapolation underestimate the  $u_i$  (see Discussion). This is why we compare four combination methods instead of adopting a single  $u_i$ -weighted mean, and why the excess-scatter term  $\xi$  below is required rather than optional. The long-averaging-time behaviour is shown by the overlapping Allan deviation of the seventeen segments spliced end-to-end on one continuous 1 s record (gaps removed), for the raw, theory and empirical scenarios, with Riley–Howe effective-degrees-of-freedom error bars (Fig. 6).

## 6.8 Statistical combination methods

The inter-segment scatter exceeds what the per-segment uncertainties  $u_i$  alone can explain, so the final ratio is obtained with a random-effect model rather than a fixed-effect average. On the same per-segment  $(R_i, u_i)$  input we compare four combination methods, both for the uncorrected baseline and for the two tidal-correction scenarios (Sec. 6.6):

- **WLS (fixed effect)**: weights  $w_i = 1/u_i^2$ , no extra scatter; the  $u_i$  are taken at face value.
- **Birge**: the  $u_i$  are rescaled by a single factor  $B = \sqrt{\chi_{\text{red}}^2}$ .
- **Mandel–Paule**: a common excess random variance  $\xi^2$  is added,  $\sigma_i^2 = u_i^2 + \xi^2$ , with  $\xi$  solved as a fixed value.

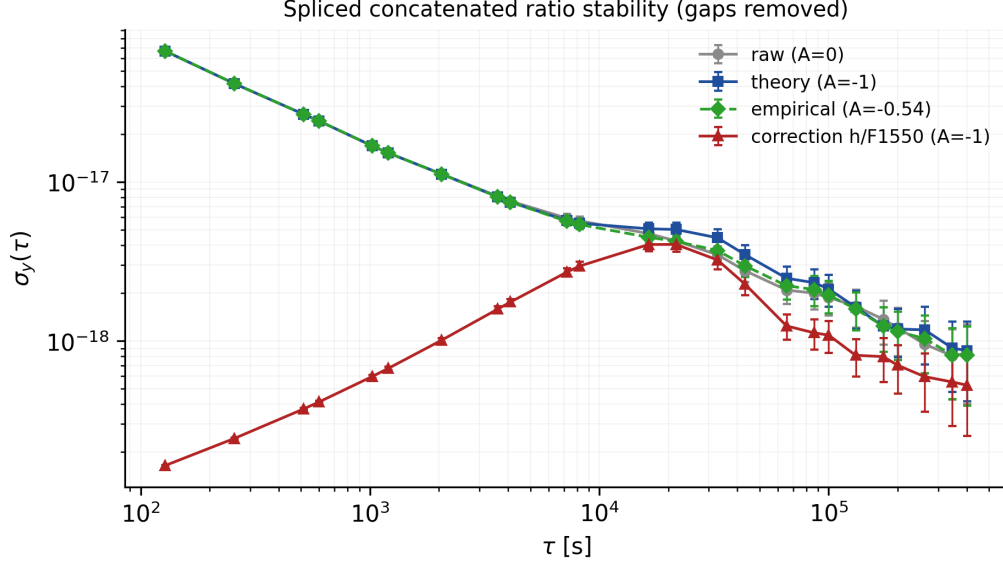


Figure 6: Spliced concatenated stability. Overlapping Allan deviation  $\sigma_y(\tau)$  of the seventeen segments placed end-to-end on one continuous 1 s record (aligned pure-data span 11.7 d, gaps removed;  $\tau$  up to 4.63 d), for the raw ( $A = 0$ ), theory ( $A = -1$ ) and empirical ( $A = -0.54$ ) scenarios, with the correction-term curve and Riley–Howe EDF error bars.

Table 6: Centre and statistical uncertainty by combination method for the uncorrected baseline, computed in this work’s own re-run of the per-segment ratios, in units of  $10^{-18}$  (deviations relative to the experiment-side WLS reference 1.207 507 039 343 337 721 3). The experiment-side analysis reports a slightly different Bayesian statistical term ( $0.75 \times 10^{-18}$ ; adopted for the headline value), reflecting its own ratio pipeline.

| Method                | Centre deviation | Statistical $u$ |
|-----------------------|------------------|-----------------|
| WLS (fixed effect)    | −0.503           | 0.275           |
| Mandel–Paule          | −0.013           | 0.755           |
| Bayesian hierarchical | −0.016           | 0.800           |

- **Bayesian hierarchical:** the same observation model as Mandel–Paule, but  $\mu$  and  $\xi$  are both inferred, so the uncertainty of  $\xi$  propagates into the final ratio through the joint posterior (sampled by MCMC).

All four methods give consistent central values, and the statistical uncertainty grows monotonically from WLS through Birge and Mandel–Paule to Bayesian (Table 6); we adopt the Bayesian result as the primary statistical uncertainty. The direction of change under the tidal correction is the same for all four methods.

## 6.9 Statistical model details

Section 6.8 introduced the four combination methods. Here we record the shared observation model and the convergence diagnostics. All four methods assume

$$R_i \sim \mathcal{N}(\mu, u_i^2 + \xi^2), \quad (10)$$

where  $\mu$  is the common true ratio and  $\xi$  the excess segment-to-segment scatter not described by the per-segment OADEV uncertainties  $u_i$ . The Bayesian hierarchical model treats both  $\mu$  and  $\xi$

as unknown and infers their joint posterior,

$$p(\mu, \xi \mid D) \propto p(D \mid \mu, \xi) p(\mu) p(\xi), \quad p(D \mid \mu, \xi) = \prod_i \frac{1}{\sqrt{2\pi(u_i^2 + \xi^2)}} e^{-\frac{(R_i - \mu)^2}{2(u_i^2 + \xi^2)}}, \quad (11)$$

with  $D = \{R_i, u_i\}$ . Because  $\xi$  is a random parameter, its posterior uncertainty propagates into the final ratio, unlike Mandel–Paule, which first fixes a point estimate of  $\xi$  and then averages.

The posterior is sampled by MCMC. Multiple independent chains mix over the full sampling range with no long-term drift or inter-chain separation, and the Gelman–Rubin convergence diagnostic,  $\hat{R}_\mu = \hat{R}_\xi = 1.000$ , confirms convergence to a common posterior. The final ratio is the posterior mean of  $\mu$ , with the posterior standard deviation as the statistical uncertainty and the 68% credible interval taken from the posterior.

## Funding

## Author contributions

## Competing interests

## Acknowledgements

## Data availability

The analysis code and derived numeric artifacts are available in the project repository at <https://github.com/hjiang555-a11y/gravitation-whsh>, released under the Apache License 2.0. The raw beat files and the professional tidal series are not redistributed here; the derived per-segment ratios, the statistical-combination results, and the tidal-correction summaries are provided as machine-readable JSON and CSV files.

## Code availability

The analysis code that produces the per-segment ratios, the statistical combinations, the tidal corrections and the leveling reduction is included in the same repository (<https://github.com/hjiang555-a11y/gravitation-whsh>) under the Apache License 2.0. The raw beat files and the professional tidal series are not redistributed; the derived products are provided as machine-readable JSON and CSV files.

## Supplementary information

Supplementary Information is provided with this manuscript, containing the per-segment data table (Supplementary Table 1), the full frequency-transfer chains and AOM offsets (Supplementary Fig. 1 and Supplementary Table 2), and the statistical-model details and MCMC convergence diagnostics (Supplementary Fig. 2).

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